Point-to-Point Response to Reviewers

Journal: *China Communications* — **Manuscript ID**: CNCOMM-2026-0222

Title: *System Relaxation for Resilient Network Governance: A Model-Driven Control Mechanism Against Structural Stress Concentration*

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Dear Associate Editor-in-Chief Prof. Ying-Chang Liang, Associate Editor, and Anonymous Reviewers,

We sincerely thank the Associate Editor-in-Chief, the Associate Editor, and the two anonymous Reviewers for their thorough reading, insightful critiques, and constructive suggestions on our manuscript (ID: **CNCOMM-2026-0222**). Your comments have helped us significantly strengthen the theoretical rigor, experimental breadth, and presentation clarity of our work.

In accordance with the journal guidelines, all modifications in the revised manuscript (`main_ccjnl.tex` / `main.tex`) are **highlighted in blue text** (`\rev{...}`). Below, we first summarize the major improvements made in this revision, followed by a detailed, point-to-point response to each comment from the Associate Editor, Reviewer 1, and Reviewer 2.

Executive Summary of Major Revisions

1. **Positioning, Research Gap, and Comparative Taxonomy (Section I, Table I, Section II-B)**:
   - Explicitly isolated the research gap between reactive queue-based load balancing, traffic-matrix-dependent IGP link-weight optimization (Fortz & Thorup), open-loop static centrality routing, and black-box AI/DRL controllers.
   - Expanded **Table I** into a 6-column feature-by-feature comparison covering *Control Signal*, *Feedback Loop*, *Adaptation Mode*, *Damping / Hysteresis*, *Complexity*, and *Key Limitation*.
2. **Rigorous Problem Formulation and Generalized Demand-/Capacity-Weighted Stress (Section III-A, Eq. (2), Eq. (5))**:
   - Introduced explicit **Assumptions 1–4** clarifying topological connectivity, observation cadence, capacity regularity, and baseline queue dynamics.
   - Formulated the **generalized demand- and capacity-weighted structural stress** $\widetilde{\kappa}_i(S; W, \mu)$ in **Eq. (2)**, which handles skewed end-to-end traffic matrices $W = [W_{st}]$ and heterogeneous node service capacities μ_i, while recovering topological betweenness centrality as a traffic-agnostic special case.
   - Added a formal **bi-criteria discrete-time optimization problem formulation (Eq. (5))** over control epochs balancing long-run peak stress and route-switching cost subject to leaky-integrator state dynamics $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha q^{(k)}$ and minimum dwell-time spacing $\tau_{m+1} - \tau_m \geq T_d$.
3. **Repaired Mathematical Proofs and Theoretical Clarifications (Section III-C/D/E, Appendices A, B, C, D, I, J)**:
   - **Lemma 1 & Appendix A**: Defined the coordinate-wise box-violation sequences $\delta^+_{k,i} := \max\{S^{(k)}_i - 1, 0\}$ and $\delta^-_{k,i} := \max\{-S^{(k)}_i, 0\}$, proving positive invariance of $X = [0,1]^N$ and exponential attraction from perturbed initial states $S^{(0)} \in [-1/\beta_I + \epsilon, \infty)^N$.

- **Proposition 1, Theorem 1, Remark 2, & Appendices B–C**: Defined the Krasovskii set-valued regularization $\mathcal{N}(S) := \bigcap_{\delta>0} \overline{\text{co}}(\text{norm}(\kappa(B_\delta(S))))$ and clarified the distinction between **global attraction on X to the compact invariant limit set** $\Omega = \bigcap_{n\geq 0} F^n(X)$ (verified via Aubin–Cellina upper-semicontinuous set-valued mappings) and convergence to a single static fixed point, explaining why multi-path routing ties can induce bounded limit cycles or invariant subsets within Ω that are temporally regularized by dwell-time gating T_d.
- **Theorem 2 & Appendix D**: Replaced the heuristic continuous-time analogy with an exact discrete-time one-step Drop-Jump inequality for the **implemented normalized feedback** $q^{(k)} = \text{norm}(\kappa^{(k)})$ and tracking energy $E_k = \|S^{(k)} - q^{(k)}\|_2^2$, proving $E_{k+1} \leq (1-\alpha)E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2 \leq (1-\alpha)E_k + \bar{\Delta}^2/\alpha$ and $\limsup_{k\to\infty} E_k \leq \bar{\Delta}^2/\alpha^2$.
- **Lemma 2 & Appendix I**: Replaced the circular conservation argument with an explicit **Route-Redistribution Budget** $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+$ and **Non-Overloading Diversion Condition** $(\kappa_j^{(m)} + B_j^{(m)} \leq \kappa_{\max}^{(m)} - \delta)$ guaranteeing strict peak stress reduction $\kappa_{\max}^{(m+1)} \leq \kappa_{\max}^{(m)} - \delta$.
- **Proposition 3 & Appendix J**: Fixed the reversed minimum inequality $(\rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i) \geq \mu_{\min}/\kappa_{\max} =: L(\kappa))$ and proved three rigorous capacity statements: (i) an unconditional improvement in the guaranteed safe arrival-rate lower bound $L(\kappa') > L(\kappa)$, (ii) exact proportionality $\rho_{\max}(\kappa')/\rho_{\max}(\kappa) = \kappa_{\max}/\kappa'_{\max}$ under homogeneous capacities $(\mu_i \equiv \mu)$ or $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max}$ under generalized capacity-normalized stress $\widetilde{\kappa}_i(S; W, \mu)$, and (iii) the exact joint (`AND`) sufficient slack conditions $(\mu_u/(\kappa_{\max} - d) \geq \rho_0$ and $\mu_j/(\kappa_j + B_j) \geq \rho_0,\ \forall j \neq u)$ under heterogeneous μ_i with unweighted centrality.

4. **Restored Missing Figures and Added Comprehensive New Experiments (Section IV-F, IV-G, IV-H, Tables III–IV, Figs. 5, 6, 9, 10, 11, 12)**:
   - **Restored Figures**: Integrated `ren_fig2_lyapunov.pdf` (**Fig. 5**, Lyapunov tracking energy decay), `ren_fig3_dwell_time.pdf` (**Fig. 6**, phase portrait and switching frequency with/without dwell time), and `sensitivity_3d_surfaces_full.pdf` (**Fig. 11**, joint 3D sensitivity over (α, β_I)).
   - **Component Ablation & Stronger Baselines (Section IV-F, Table III, Fig. 9)**: Evaluated 8 variants across 30 independent random seeds (reporting mean ± SD and normal-approximation 95% CIs $[\bar{x} \pm 1.96\, s/\sqrt{30}]$), including *Memoryless Reactive BC ($\alpha = 1$)*, *Unnormalized SRA (No* norm$(\cdot)$*)*, *SRA w/o Dwell ($T_d = 1$)*, *Static BC*, *LIRP-D*, and *Online IGP Weight Tuning (Fortz–Thorup)*, demonstrating that Full SRA $(\alpha = 0.1, \beta_I = 10, T_d = 5)$ achieves top-tier throughput (89.76 ± 0.76 pkts/step, 99.65% PDR) while cutting route switching frequency by 79.7% (20.0% vs. 98.3%).
   - **Skewed Pareto Gravity Traffic, Heterogeneous Capacities, Time-Varying Surge, and Sensing Noise (Section IV-G, Fig. 10)**: Validated generalized weighted SRA under Pareto gravity demands ($w_i \sim$ Pareto(1.5), $W_{st} \propto w_s w_t$), heterogeneous degree-correlated $\mu_i \propto \deg(i)^{0.5}$ normalized to mean 1.0, a 2× non-stationary traffic surge during $t \in [500, 1100)$ over a 1600-step horizon, and multiplicative telemetry noise $\sigma \in \{0\%, 5\%, 10\%, 20\%\}$.
   - **Dwell-Time Sensitivity, Scalability ($N = 50..500$), Runtime Benchmarks, and Real-World Backbones (Section IV-H, Table IV, Fig. 12)**: Benchmarked `NSFNET` ($N = 14$, 0.33 ms/eval), `GEANT` ($N = 22$, 0.85 ms/eval), and scale-free graphs up to $N = 500$ (850.99 ms/eval, where SRA improves seedwise relative throughput by +107.42% over Dijkstra).

5. **Dedicated Limitations Subsection & Open-Source Reproducibility (Section V-B, Section V-C)**:
    - Added **Section V-B** consolidating model assumptions, failure cases (e.g., homogeneous regular lattices/tori with $< 2\%$ gain and ultra-sparse single-bridge trees), and telemetry/scalability limitations.
    - Released all source code, LaTeX manuscripts, and multi-seed simulation pipelines in an open-source repository (https://github.com/unicorners-compin/SRA-Resilient-Network-Governance).

Response to Associate Editor (Editor: 2)

Editor Comment 1: *"First, the technical novelty should be more clearly established. The authors should explicitly distinguish SRA from existing adaptive routing and load-balancing approaches, particularly those based on dynamic link weights and centrality optimization. A clearer problem formulation and a systematic comparison with closely related methods would help strengthen the contribution."*

Author Response: We thank the Associate Editor for this crucial guidance. We have made four targeted enhancements to establish the novelty of SRA:

1. **Explicit Research Gap & Taxonomy (Section I & Table I)**: In Section I and the expanded **Table I**, we systematically contrast SRA against four representative families: (i) *Reactive Queue-Aware Routing* (backpressure/OBR), (ii) *Traffic-Matrix-Driven IGP Link-Weight Optimization* (Fortz & Thorup [26]), (iii) *Open-Loop Static Centrality Routing* [19], and (iv) *Data-Driven AI/DRL Routing* [21, 22]. Unlike reactive queue routing (which reacts only after buffer buildup and suffers from phase lag) or Fortz–Thorup IGP tuning (which requires full origin-destination traffic matrices and triggers 98.3% route switching when run online), SRA introduces a **closed-loop leaky-integrator structural state** $S_i^{(k)}$ coupled with **dwell-time gating** T_d.
2. **Expanded Related Work (Section II-B)**: We added an explicit discussion comparing SRA's node-centric structural potential and $O(N(E + N \log N))$ Brandes update against Fortz–Thorup local search over link weights and static centrality routing.
3. **Formal Problem Formulation (Section III-A, Eq. (5))**: We added a rigorous mathematical optimization formulation in **Eq. (5)** defining the bi-criteria control objective over control epochs (minimizing long-run peak structural stress and route-switching penalty subject to leaky-integrator state dynamics $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha q^{(k)}$ and minimum dwell-time spacing $\tau_{m+1} - \tau_m \geq T_d$).
4. **Empirical Comparison Against Closely Related Methods (Section IV-F, Table III, Fig. 9)**: We implemented both *Online IGP Weight Tuning (Fortz–Thorup)* and *Static Centrality Routing* alongside *LIRP-D* and *Dijkstra* across 30 seeds. As reported in **Table III**, while online IGP weight tuning achieves high throughput (90.08 ± 0.38 pkts/step), it switches routing tables at 98.3% of steps; Full SRA achieves comparable top-tier throughput (89.76 ± 0.76 pkts/step, 99.65% PDR) while reducing route switching by 79.7% (20.0% switching rate).

Editor Comment 2: *"Second, the theoretical analysis requires further clarification and justification. In particular, the convergence claims in Theorems 1 and 2 should be carefully examined, especially the distinction between attraction to a limit set and convergence to a unique equilibrium.*

The relationship between structural stress reduction and network capacity improvement, including Lemma 2 and Proposition 3, also requires stronger mathematical support."

Author Response: We completely agree and have thoroughly rewritten the theoretical statements and proofs in **Section III-C/D/E** and **Appendices A, B, C, D, I, and J**:

1. **Theorem 1 & Remark 2 (Attraction to Limit Set Ω vs. Unique Point Equilibrium, Section III-C & Appendices B–C)**: Because shortest-path routing $\mathcal{R}(G, C)$ is piecewise constant and exhibits multi-path ties at switching boundaries, its Krasovskii regularization $\mathcal{N}(S) = \bigcap_{\delta>0} \overline{\text{co}}(\text{norm}(\kappa(B_\delta(S))))$ yields an upper-semicontinuous set-valued mapping $F(S) = (1-\alpha)S + \alpha\mathcal{N}(S)$ (Aubin & Cellina [28]) rather than a contractive single-valued map. We rigorously define the global attracting invariant set on $X = [0,1]^N$ via the nested compact intersection $\Omega = \bigcap_{n\geq 0} F^n(X) \subset X$ and added **Remark 2** explicitly explaining that trajectories initialized in X converge to Ω (which may contain interior fixed points or bounded tie-breaking limit cycles/invariant subsets near switching boundaries) rather than a single static fixed point, and that dwell-time gating T_d upper-bounds the asymptotic switching frequency to at most $1/T_d$.

2. **Theorem 2 & Appendix D (Discrete-Time Drop-Jump Dissipation-Injection Bound)**: We aligned the proof in **Appendix D** directly with the implemented normalized feedback $q^{(k)} = \text{norm}(\kappa^{(k)}) \in [0,1]^N$ and tracking energy $E_k = \|S^{(k)} - q^{(k)}\|_2^2$, deriving an exact one-step discrete difference inequality $E_{k+1} \leq (1-\alpha)E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2 \leq (1-\alpha)E_k + \bar{\Delta}^2/\alpha$ without relying on heuristic continuous-time approximations.

3. **Lemma 2 & Appendix I (Non-Circular Stress Reduction Proof)**: We replaced the circular conservation argument with an explicit **Route-Redistribution Budget** $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+$ and **Non-Overloading Diversion Condition** $(\kappa_j^{(m)} + B_j^{(m)} \leq \kappa_{\max}^{(m)} - \delta)$ in **Appendix I**, proving a rigorous sufficient certificate for when a routing switch lowers the global peak stress $(\kappa_{\max}^{(m+1)} \leq \kappa_{\max}^{(m)} - \delta)$.

4. **Proposition 3 & Appendix J (Fixed Capacity Inequality & Heterogeneous Node Analysis)**: We corrected the reversed minimum inequality in the original Appendix J $(\rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i) \geq \mu_{\min}/\kappa_{\max} =: L(\kappa))$ and proved three exact statements in **Proposition 3**: (i) an unconditional increase in the guaranteed worst-case safe arrival-rate lower bound $L(\kappa') > L(\kappa)$, (ii) exact proportionality $\rho_{\max}(\kappa')/\rho_{\max}(\kappa) = \kappa_{\max}/\kappa'_{\max}$ when node service rates are homogeneous $(\mu_i \equiv \mu)$ or $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max}$ when routing uses the generalized capacity-normalized stress $\widetilde{\kappa}_i(S; W, \mu)$ and reduces $\widetilde{\kappa}_{\max}$, and (iii) the exact joint (`AND`) sufficient slack condition $(\mu_u/(\kappa_{\max} - d) \geq \rho_0$ and $\mu_j/(\kappa_j + B_j) \geq \rho_0,\ \forall j \neq u)$ under heterogeneous μ_i with unweighted centrality.

Editor Comment 3: *"Third, the experimental evaluation should be substantially strengthened. The authors are encouraged to include comprehensive ablation studies, parameter sensitivity analysis, stronger baselines, and statistical validation across multiple trials. Additional experiments under heterogeneous and time-varying traffic conditions would help demonstrate the robustness and generalizability of SRA. Computational complexity and runtime overhead should also be evaluated."*

Author Response: We have executed a comprehensive multi-seed simulation campaign across 30 random seeds and added three new experimental subsections (**Sections IV-F, IV-G, and IV-H**), two new summary tables (**Tables III and IV**), and six multi-panel figures (**Figs. 5, 6, 9, 10, 11, and 12**) covering full module ablations, stronger baselines, skewed/heterogeneous/time-varying traffic, telemetry noise injection, hyperparameter sensitivity, runtime benchmarks, and real-world backbone topologies (`NSFNET` and `GEANT`).

Editor Comment 4: *"Finally, the authors should clarify the underlying assumptions, discuss the limitations of using betweenness centrality as a stress indicator, and improve the overall presentation and reproducibility."*

Author Response: Underlying assumptions are explicitly formalized as **Assumptions 1–4** in **Section III-A**; the limitations of unweighted betweenness centrality and the generalized demand-/capacity-weighted stress $\widetilde{\kappa}_i(S; W, \mu)$ are introduced in **Eq. (2)**, tested in **Section IV-G**, and consolidated in **Section V-B**; and all code, scripts, and data are openly released in **Section V-C**.

Response to Reviewer 1

We thank Reviewer 1 for the comprehensive 34-point checklist, which provided an exceptional blueprint for elevating both the theoretical and empirical quality of our manuscript. Below we respond to each bullet point individually.

R1.1 *Strengthen the problem formulation section*: In **Section III-A**, we introduced formal system assumptions (**Assumptions 1–4**), defined the generalized demand- and capacity-weighted stress $\widetilde{\kappa}_i(S; W, \mu)$ (**Eq. (2)**), and formulated the closed-loop routing governance task as a constrained discrete-time bi-criteria optimization problem (**Eq. (5)**).

R1.2 *Provide a clearly articulated research gap statement*: Added a dedicated **Research Gap** paragraph in **Section I** (and expanded **Section II-B**) isolating the unresolved tension between reactive queue-based delay, traffic-matrix IGP route flapping, and static centrality rigidity.

R1.3 *Add a comparative summary table against closest prior methods*: Expanded **Table I** in **Section I** into a structured 6-column taxonomy comparing *Reactive Queue-Aware*, *IGP Weight Tuning (Fortz–Thorup)*, *Static Centrality Routing*, *AI/DRL Routing*, and *Proposed SRA*.

R1.4 *Expand theoretical justification of the core algorithm*: Expanded **Section III-C/D/E** and **Appendices A, B, C, D, I, and J** with complete analytical derivations for state boundedness (**Lemma 1**), compact invariant set attraction (**Proposition 1, Theorem 1, & Remark 2**), discrete-time tracking energy dissipation (**Theorem 2**), non-circular stress reduction (**Lemma 2**), and capacity lower-bound improvement (**Proposition 3**).

R1.5 *Include a full ablation study isolating each module*: Added **Section IV-F**, **Table III**, and **Fig. 9**, isolating *Memoryless Reactive BC* ($\alpha = 1$), *Unnormalized SRA (No* $\text{norm}(\cdot)$*)*, *SRA w/o Dwell* ($T_d = 1$), and *Full SRA* ($\alpha = 0.1, \beta_I = 10, T_d = 5$).

R1.6 *Introduce statistical validation across multiple trials*: All new experiments in **Sections IV-F, IV-G, and IV-H** (**Tables III–IV**, **Figs. 9, 10, 12**) are run across **30 independent random seeds**, reporting mean ± standard deviation (SD) and normal-approximation **95% confidence intervals** ($\bar{x} \pm 1.96\, s/\sqrt{30}$) in **Table III** and **Figs. 9, 10, and 12**.

R1.7 *Diversify experimental scenarios*: Added skewed Pareto gravity demands ($w_i \sim \text{Pareto}(1.5)$), heterogeneous node capacities $\mu_i \propto \deg(i)^{0.5}$, non-stationary 2× traffic surges (**Section IV-G**), scale-free and small-world networks from $N = 50$ to $N = 500$, and real-world **NSFNET** ($N = 14$) and **GEANT** ($N = 22$) backbones (**Section IV-H, Table IV**).

R1.8 *Add controlled noise injection experiments*: In **Section IV-G** and **Fig. 10(c)**, we injected multiplicative Gaussian telemetry noise $\sigma \in \{0\%, 5\%, 10\%, 20\%\}$, demonstrating that SRA's leaky integrator ($\alpha = 0.1$) maintains stable throughput (69.34 ± 12.14 pkts/step at $\sigma = 0\%$ to 69.64 ± 12.09 pkts/step at $\sigma = 20\%$, PDR $81.54\% \rightarrow 81.88\%$).

R1.9 *Clarify dataset representativeness*: Justified synthetic BA/WS/ER topologies alongside real-world **NSFNET** and **GEANT** backbones and heavy-tailed Pareto gravity traffic matrices in **Sections IV-A, IV-G, and IV-H**.

R1.10 *Provide computational complexity analysis*: Explicitly analyzed per-epoch time complexity $O(N)$ (within frozen dwell steps where cached paths and stresses are reused) and $O(N(E + N\log N))$ (at block boundaries where Brandes' algorithm is executed, amortized to $O(N(E + N\log N)/T_d)$ per step) as well as working memory $O(N + E)$ and routing-table storage $O(N^2)$ in **Section III-B (Algorithm 1)** and **Section IV-H**.

R1.11 *Report runtime benchmarks on specified hardware*: Benchmarked single-thread wall-clock execution time on an **Intel Xeon Gold 6248R (3.00 GHz)** server in **Table IV** and **Fig. 12(b)** (0.33 ms on NSFNET, 0.85 ms on GEANT, 31.17 ms at $N = 100$, and 850.99 ms at $N = 500$).

R1.12 *Improve parameter selection justification*: Provided a control-theoretic selection rationale linking $\alpha \in [0.08, 0.25]$ to the discrete relaxation time constant $\tau_{\text{relax}} = -1/\ln(1-\alpha) \approx 1/\alpha$, $\beta_I \in [8, 20]$ to network hop diameter, and $T_d = 5$ to τ_{relax} in **Sections III-D and IV-H**.

R1.13 *Include hyperparameter sensitivity plots*: Restored the 3D joint sensitivity surfaces over (α, β_I) in **Fig. 11** and added the $T_d \in \{1..10\}$ sensitivity plot in **Fig. 12(a)**.

R1.14 *Clarify boundary conditions and assumptions*: Stated explicitly as **Assumptions 1–4** in **Section III-A**.

R1.15 *Expand discussion of failure cases*: Added **Section V-B** analyzing explicit boundary/failure cases: (i) homogeneous regular lattices/tori (where $\kappa_{\max} = \bar{\kappa}$ initially, yielding $< 2\%$ gain) and (ii) ultra-sparse single-bridge/cut-vertex trees (where $\Delta^-_{v_{\text{cut}}} = 0$ and zero bypass paths exist).

R1.16 *Include visual comparison under challenging scenarios*: Added **Fig. 10** comparing *Dijkstra*, *Topology-Only SRA*, and *Generalized Weighted SRA* under skewed Pareto gravity demand, a $2\times$ traffic surge ($t \in [500, 1100)$), and $\sigma \in [0\%, 20\%]$ sensing noise.

R1.17 *Improve figure readability and annotation quality*: Regenerated multi-panel figures (**Figs. 9, 10, 12**) with clear legends, 95% CI error bars, and annotated surge intervals, and restored vector PDFs for **Figs. 5, 6, and 11**.

R1.18 *Reorganize methodology for logical flow*: Structured **Section III** sequentially from Assumptions, Stress Metrics & Optimization Formulation → Algorithm 1 → Set-Valued Stability Proofs → Dwell-Time Regularization → Capacity Analysis.

R1.19 *Reduce repetitive narrative in discussion*: Streamlined **Section V-A** to eliminate repetition and focus on control-theoretic synthesis.

R1.20 *Condense conclusion to focus on quantified findings*: Rewrote **Section V-C** around concrete quantitative findings (83.76% peak reduction, +29.79% throughput gain at $N = 100$, +107.42% at $N = 500$, 79.7% switching reduction, and +20.28% under skewed Pareto traffic).

R1.21 *Provide clearer algorithm pseudocode*: Updated **Algorithm 1** with explicit dwell-time block gating τ_{dwell}, path/stress caching during cooling steps, generalized stress option $\widetilde{\kappa}_i$, and complexity analysis.

R1.22 *Explicitly state limitations in a dedicated subsection*: Created **Section V-B (*Explicit Assumptions, Failure Cases, and Limitations*)**.

R1.23 *Compare against stronger baselines*: Added **Online IGP Weight Tuning (Fortz–Thorup)** and **Static Centrality Routing** alongside **LIRP-D** and **Dijkstra** in **Table III** and **Fig. 9**.

R1.24 *Explain metric selection rationale*: Explained in **Sections IV-A and IV-F** how Throughput, PDR, Mean/P95 Latency, Peak Stress Reduction, Fairness, Switching Rate, and Runtime jointly evaluate data-plane performance and control-plane stability.

R1.25 *Include additional evaluation metrics*: Added **P95 Tail Latency**, **PDR (%)**, **Route Switching Rate** $\bar{f}_{\mathbf{sw}}$, and **Per-Evaluation Runtime (ms)** in **Tables III–IV**, and reported **Max Generalized Stress** $\max_i \widetilde{\kappa}_i$ in **Section IV-G** and **Fig. 10(a)**.

R1.26 *Discuss scalability to larger or more complex inputs*: Evaluated $N \in \{50, 100, 200, 300, 500\}$ in **Table IV** and **Fig. 12(b)–(c)**, showing +107.42% seedwise relative throughput gain at $N = 500$.

R1.27 *Clarify training and evaluation protocol details*: Documented all simulation horizons, warm-up steps, traffic models, and fixed hyperparameters in **Section IV-A** and **Table II**.

R1.28 *Improve mathematical notation consistency*: Unified k (epoch), E (edge count), M (polyhedral routing strata count), $S_i^{(k)}$ (normalized state), $q_i^{(k)} = \mathrm{norm}(\kappa_i^{(k)})$ (normalized instantaneous stress), $E_k = \|S^{(k)} - q^{(k)}\|_2^2$ (tracking energy), μ_i (service rate), and T_d (dwell time) across all sections and appendices.

R1.29 *Strengthen connection between theory and experiments*: Explicitly linked **Theorem 1** to **Figs. 6 & 12(a)**, **Theorem 2** to **Fig. 5**, and **Lemma 2 / Proposition 3** to **Table III & Figs. 9–10**.

R1.30 *Provide ablation on preprocessing or auxiliary steps*: Evaluated max-normalization ($\mathrm{norm}(\cdot)$) vs. unnormalized SRA (raw $\kappa^{(k)}/N^2$) in **Section IV-F** and **Table III**.

R1.31 *Address potential data leakage or evaluation bias*: Confirmed that a single fixed parameter tuple ($\alpha = 0.1, \beta_I = 10, T_d = 5$) is applied across all new ablation, heterogeneous, and scalability benchmarks without per-test retuning, evaluated over 30 independent test seeds.

R1.32 *Improve figure captions to be self-contained*: Expanded captions of **Figs. 5–12** and **Tables I–IV** with full experimental context and takeaways.

R1.33 *Update bibliography with recent relevant literature*: Added Fortz & Thorup [26], Brandes [27], and Aubin & Cellina [28] alongside recent IEEE JSAC/TCOM/TNSE references.

R1.34 *Provide code or implementation details for reproducibility*: Provided the open-source GitHub repository in **Section V-C** (https://github.com/unicorners-compin/SRA-Resilient-Network-Governance).

Response to Reviewer 2

We sincerely thank Reviewer 2 for the exceptionally detailed, constructive, and mathematically rigorous review. Your careful inspection of the proofs in Appendices A, C, D, I, and J and the missing figure references in Section IV was invaluable. Below we address each of your five major concerns in detail.

Reviewer 2 — Comment 1: Novelty, Related Work Distinction, and Baselines

Comment: *"1. Novelty and Distinction from Existing Adaptive Routing / Centrality Routing: The manuscript states that existing adaptive routing methods are reactive and local, while SRA*

is proactive and topological. However, using node/link centrality (such as betweenness centrality) or dynamic link weights to balance load is a well-studied topic in traffic engineering (e.g., IGP weight optimization by Fortz & Thorup, centrality-aware routing in complex networks, and exponentially weighted moving average link costs). The authors need to more clearly articulate what is fundamentally new in SRA beyond applying an EWMA filter (α) and a dwell-time threshold (T_d) to normalized betweenness centrality. Furthermore, in the experiments, SRA is only compared against static Dijkstra and a single heuristic baseline (LIRP-D). Comparing against a static centrality-weighted routing scheme or a standard congestion/weight-smoothing adaptive routing baseline is necessary to isolate where the gain comes from."

Author Response: We thank Reviewer 2 for raising this central question. We have addressed both the conceptual distinction and the experimental isolation of gains:

1. **What is Fundamentally New in SRA (Section I, Table I, Section II-B, Section III-C/D)**:
   - Unlike **Static Centrality Routing** (which computes betweenness centrality once on the initial unweighted graph and freezes node costs), open-loop one-shot penalty cannot adapt to topological changes or non-stationary traffic shifts, and risks over-penalizing primary hubs onto secondary nodes without feedback correction.
   - Unlike **Online IGP Weight Tuning (Fortz & Thorup [26])** or **Memoryless Reactive Centrality Routing ($\alpha = 1$)**, directly feeding instantaneous topological or utilization metrics back into shortest-path routing creates a **discontinuous piecewise-constant closed-loop map** that suffers from severe route flapping (chattering across tie-breaking boundaries at 98.3% of steps).
   - SRA's core contribution is the **joint dynamical coupling** of (i) a continuous leaky-integrator structural stress state $S_i^{(k)}$ whose Krasovskii set-valued regularization $F(S) = (1-\alpha)S + \alpha\mathcal{N}(S)$ admits a provably compact global attractor Ω on $X = [0,1]^N$ (**Theorem 1**) and bounded discrete-time tracking energy (**Theorem 2**), and (ii) a **dwell-time block gate** T_d matched to the relaxation time constant $\tau_{\text{relax}} = -1/\ln(1-\alpha) \approx 1/\alpha$ that suppresses boundary chattering on Ω while preserving closed-loop bottleneck evasion (**Lemma 2, Proposition 3**).
2. **Experimental Isolation Against Static Centrality, Memoryless Reactive BC, and Online IGP Weight Tuning (Section IV-F, Table III, Fig. 9)**: We implemented and evaluated all requested baselines and ablations across 30 seeds on BA scale-free networks ($N = 100, \rho = 0.9$):
   - **Memoryless Reactive BC ($\alpha = 1, T_d = 1$)**: Without leaky-integrator smoothing, routing paths oscillate violently every step (98.3% switching rate), *amplifying* peak centrality by 31.04% and degrading throughput to 53.39 ± 5.88 pkts/step (worse than static Dijkstra's 68.88 ± 5.57 pkts/step).
   - **Static Betweenness Centrality Routing**: Eliminates continuous route switching (1.7%) and improves stationary uniform-load throughput over Dijkstra (89.96 ± 0.51 pkts/step), but is strictly open-loop and cannot adapt when traffic demands or network states change over time.
   - **Online IGP Weight Tuning (Fortz–Thorup)**: Achieves high throughput (90.08 ± 0.38 pkts/step) and peak stress reduction (77.03%), but requires continuous route-table switching at 98.3% **of control epochs**.
   - **SRA w/o Dwell ($T_d = 1$) vs. Full SRA ($\alpha = 0.1, \beta_I = 10, T_d = 5$)**: Adding the leaky integrator ($\alpha = 0.1$) stabilizes reactive BC (53.39 → 90.08 pkts/step), and adding

the dwell-time gate ($T_d = 5$) cuts route switching frequency by 79.7% (98.3% $\rightarrow$ 20.0%) while retaining 99.65% PDR (89.76 ± 0.76 pkts/step).

Reviewer 2 — Comment 2: Mathematical Rigor in Proofs (Theorems 1 & 2, Lemma 2, Proposition 3, Supplement Sections S-A, S-C, S-D, S-I, S-J)

Comment: *"2. Mathematical Rigor and Gaps in Theoretical Proofs: Several proofs in the appendices contain mathematical gaps or imprecise statements that need careful revision: (a) Theorem 1 and Appendix C (unique equilibrium vs. compact invariant set Ω, undefined $\delta^+_{k,i}, \delta^-_{k,i}$ in Appendix A); (b) Theorem 2 and Appendix D (discontinuous $\kappa(S)$, heuristic continuous-time analogy $\dot{V}$, unnormalized $\kappa(S)$ vs. $\mathrm{norm}(\kappa)$); (c) Lemma 2 (Appendix I) and Proposition 3 (Appendix J) (conservation law $\sum_i \kappa'_i = \sum_i \kappa_i$, condition for $\max_{j \notin \mathcal{B}} \kappa'_j < \kappa_{\max}$, reversed minimum inequality $\min_i(\mu_i/\kappa_i) \le \mu_{\min}/\kappa_{\max}$, and OR vs. AND condition)."*

Author Response: We are deeply grateful to Reviewer 2 for pinpointing each of these exact mathematical issues. We have completely rewritten the corresponding theorem statements and proof sketches in **Section III-C/D/E** and the full formal proofs in **Supplementary Material Sections S-A, S-B, S-C, S-D, S-I, and S-J**:

1. **Fixed Lemma 1 & Supplement Section S-A (Explicit Definition of $\delta^+_{k,i}, \delta^-_{k,i}$)**: Defined the coordinate-wise box-violation excesses $\delta^+_{k,i} := \max\{S_i^{(k)} - 1,\, 0\}$ and $\delta^-_{k,i} := \max\{-S_i^{(k)},\, 0\}$ satisfying $\delta^{\pm}_{k+1,i} \le (1-\alpha)\delta^{\pm}_{k,i}$, proving positive invariance of $X = [0,1]^N$ and exponential attraction from perturbed initial states $S^{(0)} \in [-1/\beta_I + \epsilon, \infty)^N$.

2. **Fixed Proposition 1, Theorem 1, Remark 2, & Supplement Sections S-B–S-C (Attraction to Compact Invariant Limit Set Ω vs. Unique Point Equilibrium)**: Defined the Krasovskii regularization $\mathcal{N}(S) := \bigcap_{\delta>0} \overline{\mathrm{co}}(\mathrm{norm}(\kappa(B_\delta(S))))$ and revised **Theorem 1** to prove global attraction on X to the non-empty, compact, invariant limit set $\Omega = \bigcap_{n=0}^{\infty} F^n(X) \subset X$ ($\lim_{k\to\infty} \mathrm{dist}(S^{(k)}, \Omega) = 0$) via Aubin–Cellina upper-semicontinuous set-valued mappings [28], and added **Remark 2** distinguishing attraction to Ω from convergence to a unique point equilibrium.

3. **Fixed Theorem 2 & Supplement Section S-D (Normalized Feedback $q^{(k)} = \mathrm{norm}(\kappa^{(k)})$ and Exact Discrete-Time Step Bound)**: Unified **Theorem 2** and **Supplement Section S-D** with the implemented normalized feedback $q^{(k)} = \mathrm{norm}(\kappa^{(k)}) \in [0,1]^N$ and tracking energy $E_k = \|S^{(k)} - q^{(k)}\|_2^2$. Using the exact discrete identity $S^{(k+1)} - q^{(k+1)} = (1-\alpha)(S^{(k)} - q^{(k)}) + (q^{(k)} - q^{(k+1)})$ and Weighted Young's inequality with $\lambda = \frac{\alpha}{1-\alpha}$, we proved:

$$E_{k+1} \le (1-\alpha)E_k + \frac{1}{\alpha}\left\|q^{(k)} - q^{(k+1)}\right\|_2^2 \le (1-\alpha)E_k + \frac{\bar{\Delta}^2}{\alpha} \implies \limsup_{k\to\infty} E_k \le \frac{\bar{\Delta}^2}{\alpha^2}.$$

4. **Fixed Lemma 2 & Supplement Section S-I (Non-Circular Route-Redistribution Budget $B_j^{(m)}$)**: Removed the circular conservation assumption $\sum_i \kappa'_i = \sum_i \kappa_i$ and defined each non-peak node's route-redistribution budget $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+ \le \sum_{(s,t)\in\mathcal{D}_{\mathrm{in}}^{(m)}(j)} W_{st}$. Under positive load shedding at the unique pre-switch peak u ($\Delta_u^- := \kappa_u^{(m)} - \kappa_u^{(m+1)} \ge \delta > 0$) and the explicit **Non-Overloading Diversion Condition** ($\kappa_j^{(m)} + B_j^{(m)} \le \kappa_{\max}^{(m)} - \delta,\ \forall j \in V \setminus \{u\}$), we proved $\kappa_{\max}^{(m+1)} \le \kappa_{\max}^{(m)} - \delta$.

5. **Fixed Proposition 3 & Supplement Section S-J (Corrected Minimum Inequality and `AND` Sufficient Slack Condition)**: Corrected the inequality to $\rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i) \ge \mu_{\min}/\kappa_{\max} =: L(\kappa)$ (where ρ is the per-OD arrival-rate multiplier), and proved: (i) unconditional improvement of the guaranteed safe arrival-rate lower bound

$L(\kappa') > L(\kappa)$; (ii) exact proportionality $\rho_{\max}(\kappa')/\rho_{\max}(\kappa) = \kappa_{\max}/\kappa'_{\max}$ under homogeneous $\mu_i \equiv \mu$, and $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max}$ under generalized capacity-normalized stress $\widetilde{\kappa}_i(S; W, \mu)$ whenever $\widetilde{\kappa}'_{\max} < \widetilde{\kappa}_{\max}$; and (iii) the exact joint (`AND`) sufficient condition $\mu_u/(\kappa_{\max} - d) \geq \rho_0$ and $\mu_j/(\kappa_j + B_j) \geq \rho_0$ ($\forall j \in V \setminus \{u\}$, with $d := \Delta_u^- \geq \delta > 0$) when using unweighted centrality under heterogeneous μ_i.

Reviewer 2 — Comment 3: Betweenness Centrality vs. Non-Uniform Traffic & Heterogeneous Capacities

Comment: *"3. Betweenness Centrality as a Proxy for Stress under Non-Uniform Traffic: ... How does SRA perform when traffic demands are non-uniform or time-varying, or when node capacities μ_i are heterogeneous? Discussing Demand-Weighted Betweenness or Capacity-Normalized Stress—and testing SRA under non-uniform/time-varying traffic—would make the framework much more convincing."*

Author Response: We introduced the **generalized demand- and capacity-weighted stress** $\widetilde{\kappa}_i(S; W, \mu) = \frac{1}{\mu_i} \sum_{s \neq i \neq t} W_{st} \frac{\sigma_{st}(i;S)}{\sigma_{st}(S)}$ in **Section III-A (Eq. (2))** and evaluated it in **Section IV-G (Fig. 10)** under heavy-tailed Pareto gravity traffic ($w_i \sim \text{Pareto}(1.5)$, $W_{st} \propto w_s w_t$), heterogeneous degree-correlated $\mu_i \propto \deg(i)^{0.5}$ normalized to mean 1.0, and a 2× mid-run traffic surge during $t \in [500, 1100)$ over 1600 steps. Generalized Weighted SRA reduces peak generalized stress by 55.6% (from 2364.8 to 1051.1), improves throughput by +20.28% over Dijkstra (69.34 ± 12.14 vs. 57.65 ± 8.97 pkts/step, PDR 81.54% vs. 67.79%), reduces P95 tail latency from 12.93 to 9.83 steps, and outperforms Topology-Only SRA (68.91 ± 12.04 pkts/step).

Reviewer 2 — Comment 4: Missing Figures, Dwell-Time Sensitivity, Ablation, and Scalability/Runtime

Comment: *"4. Experimental Completeness and Missing Visual Evidence: Restore 'ren_fig2_lyapunov.pdf', 'ren_fig3_dwell_time.pdf', and 'sensitivity_3d_surfaces_full.pdf'; add sensitivity analysis for T_d and ablation of $T_d = 1$ and $\alpha = 1$; evaluate runtime and scalability from $N = 50$ to 500 and on NSFNET/GEANT."*

Author Response:

1. **Restored All Three Omitted Figures**: Restored as **Fig. 5** (`ren_fig2_lyapunov.pdf`), **Fig. 6** (`ren_fig3_dwell_time.pdf`), and **Fig. 11** (`sensitivity_3d_surfaces_full.pdf`).
2. **Dwell-Time Sensitivity & Ablation**: Added in **Table III**, **Fig. 9**, and **Fig. 12(a)** across $T_d \in \{1, 2, 3, 5, 8, 10\}$ and $\alpha = 1$.
3. **Scalability ($N = 50..500$) & Real Backbones (`NSFNET`, `GEANT`)**: Added in **Section IV-H**, **Table IV**, and **Fig. 12(b)–(c)**, reporting both wall-clock runtime (0.33 ms on NSFNET up to 850.99 ms at $N = 500$) and seedwise relative throughput scaling (+107.42% gain at $N = 500$).

Reviewer 2 — Comment 5: Minor Formatting and Parameter Clarification Issues

Comment: *"5. Minor Presentation and Formatting Issues: Remove '[cite: 276, 277]' from Figure 5 caption; clarify $C_{\max} = 50$ vs. $\mu = 2.0$ in Table II."*

Author Response: Removed the '[cite: 276, 277]' artifact from the congestion transition figure caption (now **Fig. 7**), and explicitly distinguished Node Buffer Capacity $B_{\max} = 50$ packets from Node Service Rate $\mu = 2.0$ packets/step in **Table II**.

We thank the Associate Editor-in-Chief, the Associate Editor, and the two Reviewers again for their constructive feedback and time spent reviewing our work.